Seat No._____

GANPAT UNIVERSITY

B.Sc.IT(DS) SEM-II EXAMINATION

U72A1IP2: INTRODUCTION TO PROGRAMMING-II

Time: 3 Hours] April - June 2023 **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

Q.1 Answer the following: (Any Five out of Seven) **(30)**

A) Explain String Handling functions with example.

B) Discuss Two-Dimensional Array with Example

C) What is UDF? Why we need a User Define Function?

D) Explain Nesting of functions with example.

E) Discuss structure within structure with example.

F) Explain the difference between call by value and call by reference with example.

G) Explain Dynamic Memory Allocation in detail.

SECTION-II

Q.2 Answer the following: (any two) **(12)**

A) Write a short note on Table of String.

B) Explain run time initialization of One Dimensional Array with example.

C) What is Recursion? Explain it with example.

Q.3 Answer the following: (any two) **(12)**

A) Write a short note on Union.

B) What is Pointer? Discuss advantages of pointer.

C) Discuss fprintf() and fscanf() function with example.

Q.4 Answer the following: (any one) **(06)**

A) Explain File Inclusion in detail.

B) Write a short note on Preprocessor.

END OF PAPER

Seat No._____

GANPAT UNIVERSITY
B.Sc. IT(DS) SEM-II EXAMINATION

U72A2DA2: DATA ANALYSIS-II
April - June 2023

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. **Answer the following: (Any Five)** **(30)**

1. What are financial functions? Explain PMT, IPMT, PPMT, NPER, and RATE.
2. Distinguish between Database functions and other functions in Excel. Explain any 5 Database functions.
3. What are Sparklines? Give an example of summarizing data using Sparklines.
4. Compare LOOKUP, VLOOKUP, and HLOOKUP functions in Excel with an appropriate example.
5. What is Solver? Explain Solver in detail.
6. How can Macros be used to automate repetitive tasks?
7. How to create Hyperlinks in worksheets?

SECTION-II

2. **Answer the following: (Any Two)** **(12)**

1. Explain any seven Statistical functions with examples in detail.
2. Describe Line and Bar chart in detail.
3. Demonstrate ways in which you can fine-tune your charts. Describe Doughnut and Pie chart with an appropriate example.

3. **Answer the following: (Any Two)** **(12)**

1. Explain What-if-analysis in detail. Discuss Scenario Manager with an appropriate example.
2. What is Goal Seek in Excel, and how can it be used to find the result?
3. Explain the use of Pivot tables. How to create a Pivot table using an external data source?

4. **Answer the following: (Any One)** **(06)**

1. Explain various ways in which we can protect worksheets and workbooks.
2. Define Headers and Footers. Write the steps to add a Header and Footer to your Worksheets.

END OF PAPER

Seat No._____

GANPAT UNIVERSITY

B.Sc. IT (DATA SCIENCE) SEM-II EXAMINATION

April - June 2023

U72A3DMS: DATABASE MANAGEMENT SYSTEM

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks.
2. Each section should be written in a separate answer book.
3. Be precise and to the point in your answer.

SECTION-I

1. Answer the following: (Any Five) **(30)**

1. Explain Responsibilities of Database Administrator.
2. Write down Dr. E.F. Codd's 12 Rules for RDBMS.
3. Explain B/C Normal From with example.
4. What is the constraints in DBMS? Explain types of constraints in detail.
5. Explain the logical operators in DBMS with example.
6. Explain GROUP BY ROLLUP and GROUP BY CUBE in detail.
7. Explain index and view in detail.

SECTION-II

2. Answer the following: (Any Two) **(12)**

1. Difference Between Hierarchical Network and Relational Database Model.
2. Explain E-R diagram.
3. What are Functional dependencies in DBMS? Explain types of Functional dependencies in detail.

3. Answer the following: (Any Two) **(12)**

1. Explain in brief usage with syntax of alter, truncate, and drop commands.
2. Explain types of joins.
3. How are set operators differ from joins? Explain set operators in detail.

4. Answer the following: (Any One) **(06)**

1. Explain Grant and Revoke features of DBMS with example.
2. Design normalized database design with ER relationship of College Management System.

END OF PAPER

Seat No._____

GANPAT UNIVERSITY
B.Sc.IT (DS) SEM-II EXAMINATION
April - June 2023
U72A4COA: COMPUTER ORGANIZATION AND ARCHITECTURE

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

1. Answer the following: (Any Five) **(30)**

1. $(65)_{10}$ = ($)_2$ And ($1011011011)_2$ = ($)_{16}$
2. Explain the Error detection methods in brief with example.
3. (3) + (-5) Perform the Binary Addition using 2's Compliment method.
4. State the difference between CISC & RISC computer architectures.
5. Elaborate the Instruction execution cycle with figure.
6. Describe the 8085 Microprocessor pin diagram with figure.
7. Explain the Arithmetic Instructions with example

SECTION-II

2. Answer the following: (Any Two) **(12)**

1. Explain Excess – 3 Code in Detail.
2. Do as Directed.
   $(330)_8$ = ($)_{10}$ And $(A56B)_{16}$ = ($)_{10}$
3. State the difference between Microcontroller & Microprocessor

3. Answer the following: (Any Two) **(12)**

1. Elaborate the Pipelining in brief.
2. Detail of the Addressing techniques with example.
3. Explain the 8085 Microprocessor architecture with figure.

4. Answer the following: (Any One) **(06)**

1. Illustrate the Logic Instructions with example.
2. Describe the Assembly Programming Instructions with example.

END OF PAPER

Seat No._____

GANPAT UNIVERSITY
B.Sc.IT (DATA SCIENCE) SEM-II EXAMINATION

U72A5BS BASIC STATISTICS
April - June 2023

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer
4. Simple calculator is allowed

SECTION-I

1. Answer the following: (Any Five) **(30)**

1. List out merits and demerits of Arithmetic Mean.
2. Find the median of the following data

Class	0-30	30-60	60-90	90-120	120-150	150-180
Frequency	8	13	22	27	18	7

3. Calculate standard deviation from the following distribution.

Age (year)	20-25	25-30	30-35	35-40	40-45	45-50
No. of Persons	170	110	80	45	40	35

4. For the following distribution determine the mean deviation and coefficient of mean deviation.

x	10	11	12	13	14
f	3	12	18	12	3

5. If a fair dice is rolled twice, observe the numbers that face up. Find the conditional probability that the sum of the number is 7, given that the first number is 2.
6. Compute the correlation coefficient for the following data and interpret it.

Marks in English	4	5	9	14	18	22	24
Marks in Statistics	16	22	11	16	7	3	17

7. A card is drawn is at random from a pack of 52cards. What is the probability that the card is a spade or king?

SECTION-II

2. Answer the following: (Any Two) **(12)**

1. Explain geometric mean and Harmonic mean in details.
2. You are given the frequency distribution of sales per day

Sales Volume	53-56	57-60	61-64	65-68	69-72	72 and above
Number of days	2	4	5	4	4	1

Determine the mode of sales distribution of the units of item during the 20 days period.

3. The following is the distribution of wages of some workers . Determine quartile deviation and coefficient of quartile deviation.

Wages (in Rs)	20	32	61	75	82	95
No. of workers	2	4	7	5	4	2

3. Answer the following: (Any Two) **(12)**

1. Find the correlation coefficient between the serum diastolic blood pressure and serum cholesterol levels of 10 persons.

Cholesterol	307	259	341	317	274	416	267	320	274	336
Diastolic B.P	80	75	90	74	75	110	70	85	88	78

2. List out the properties of regression coefficient.
3. Obtain the regression line x on y from the following data. Also compute x when $y = 35$.

Sales (x)	190	240	250	300	310	335	300
Advertising Expenditure(y)	5	10	15	20	20	30	30

4. Answer the following: (Any One) **(06)**

1. Explain Bayes' theorem for three events.
2. In a bolt factory, three machines, A, B and C manufacture 30%, 45% and 25 % of the total product respectively. Of these outputs 4%, 3% and 2% respectively, are defective bolts. A bolt is picked up at random and found to be defective. What is the probability that the bolt came from machine B?

END OF PAPER

Enrollment No._____

GANPAT UNIVERSITY

B.Sc. (DATA SCIENCE) SEM-II EXAMINATION

U72B6CS2: LANGUAGE AND COMMUNICATION SKILLS – II

April - June 2023

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

SECTION-I

Q.1 **Answer the following: (Any Five)** **(30)**

A. Discuss the following terms:

Synonyms, Antonyms, Prefixes, and Suffixes.

B. Write expressions for the following situations:

a) Asking for the direction

b) Inviting a friend for a birthday party

c) Expressing gratitude for receiving the DG award in the university

C. What do you mean by presentation skill? Discuss various characteristics to deliver a presentation effectively.

D. What is an advertisement? Explain the types of it with examples.

E. Discuss various objectives of Group Discussion.

F. Draft a NOTICE for the residence of your society regarding inauguration of library for all the residents. You may include details like, time, location, chief guest, refreshment, etc.

G. What is a press release? Discuss various types of it.

SECTION-II

Q.2 **Answer the following: (Any Two)** **(12)**

A. **Do as directed:**

I. **Use the suffix and make a word:** '-ness'

II. **Use the prefix and make a word:** 'ir-'

III. **Give Synonyms for the word:** 'Dias'

IV. **Give Antonyms for the word:** 'Top'

V. **Give one word substitution**: One who can speak multiple languages

VI. **Identify errors and rewrite the sentence:** They were(a)/ played cricket (b)/ when we(c)/ reached their home. (d)

B. **Use Appropriate forms of verb and fill in the blanks:**

I. Varun _____ speak French fluently. **(Can/May)**

II. My grandparents gave me an ____ of Rs.5000/-. **(envelop/envelope)**

III. Neha and Rekha _____ my siblings. **(Use appropriate form of 'to be')**

IV. Karan ____already submitted the assignment before the teacher asked him. **(had/have)**

V. Students _____ a disc in a laptop. **(added/inserted)**

VI. If Neha performed well, she ____get selected for the finale. **(will/would)**

C. Make an announcement regarding the seminar that your department is going to organize on the topic, 'Career in Data Science'. You can include details like, date, time, location, registration, contact information and prize, etc.

Q.3 **Answer the following: (Any Two)** **(12)**

A. Write a letter to Mr. Mohan Raj, Admin Officer, CRC University, Chennai, to inquire about the masters in Data Science program. You can include details like, hostel availability, mess, placement ratio, teaching and learning environment and other facilities. Use block layout.

B. What are the boredom factors in presentation skills? Discuss its remedies.

C. Prepare agenda and minutes of meeting for the meeting conducted regarding celebration of Annual Day. You can include details like; committee, cultural programs, chief guest, refreshment and other required information.

Q.4 **Answer the following: (Any One)** **(06)**

A. What are the prerequisites of Group Discussion?

B. Discuss Do's and Don'ts of Group Discussion.

END OF PAPER